

Probability

$$P(\text{this is bollocks}) = 1$$

Permutations and Combinations

Permutations count distinct objects in an **ordered** arrangement.

- An r -permutation of a set of n elements is an ordered arrangement of r unique elements from the set
- The number of r -permutations of a set of n elements is given by:

$$P(n, r) = \frac{n!}{(n-r)!}$$

Permutations: $1, 2, 3 ! = 3, 2, 1$

If a set consists of **indistinguishable** objects of k types:

$$P(n; n_1, \dots, n_k) = \frac{n!}{n_1! n_2! \dots n_k!}$$

Where $n_1, \dots, n_k$ represent the number of occurrences of each type within the set.

Combinations count distinct objects in an **unordered** arrangement

- An r -combination is an unordered arrangement of r elements from a set
- The number of r -combinations of a set with n distinct elements is given by:

$$C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)! r!}$$

Combinations: $1, 2, 3$ is the same as $3, 2, 1$

Basic Probability

The **sample space** S is the set of all possible outcomes of an experiment.

- An **event** or **outcome** is a subset of S .

For finite S , the probability of an **event**, E occurring is:

$$P(E) = \frac{|E|}{|S|} \rightarrow \begin{array}{l} \text{no. of outcomes where } E \text{ occurs} \Rightarrow \text{a subset of } S. \\ \text{total number of outcomes in } S \end{array}$$

Basic Rules:

- $P(S) = 1$
- $0 \leq P(E) \leq 1$
- $P(\bar{E}) = 1 - P(E)$
- $P(\emptyset) = 0$
- $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

For two events A and B :

- The probability both A and B occur: $P(A \cap B)$
- The probability either A or B occur: $P(A \cup B)$
- The probability of A occurring and B not occurring: $P(A \cap \bar{B}) = P(A - B)$

Conditional Probability

The probability of an event A occurring, given that another event B has already happened is given by:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

If A and B independent, then $P(A|B) = P(A)$, Since the occurrence of B does not affect the probability of A .

Substituting in the above formula gives:

$$P(A) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A)P(B)$$

for independent events.

Note: mutually exclusive events are always dependent

Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Labels: $P(B|A)$ is likelihood, $P(A)$ is prior, $P(A|B)$ is posterior, $P(B)$ is evidence.

For n mutually exclusive events $A_1, \dots, A_n$ such that $\sum_{i=1}^n A_i = S$ (equivalently: $\sum_{i=1}^n P(A_i) = 1$, i.e. exactly one A_i must occur), then,

For any event B :

$$P(A_k|B) = \frac{P(A_k) \times P(B|A_k)}{\sum_{j=1}^n P(A_j) \times P(B|A_j)}$$

Labels: $\sum_{i=1}^n A_i = S$ is Sample Space.

Discrete Random Variables

A **random variable** is a function that assigns a number to each possible outcome of an experiment.
 $\Rightarrow$ Eg. we can use a **random variable** to represent the number of heads in n coin flips. In this case the number assigned to each possible outcome represents the number of heads.

Such a random variable is **discrete** if it has countably many possible values.

The **probability distribution** of a **random variable** describes the probabilities associated with each possible outcome of the variable.

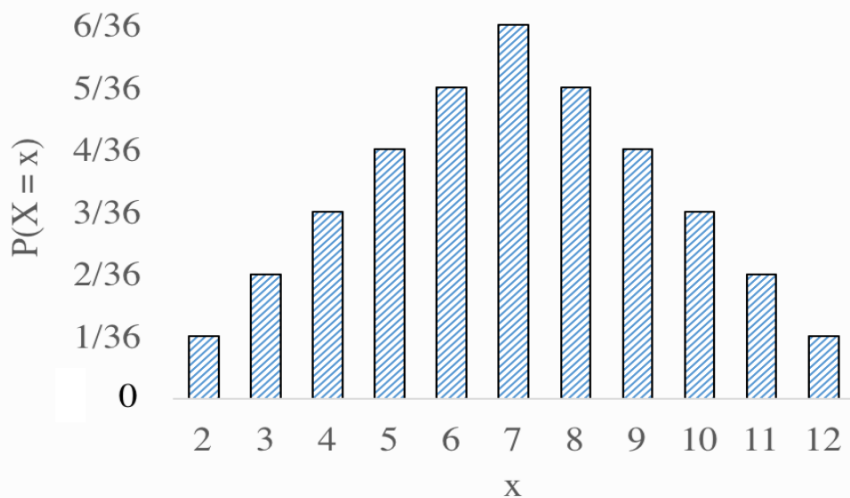
The **Probability Mass Function (PMF)** of a **discrete random variable** X with possible values $x_1, \dots, x_n$ is the function $f(x_k) = P(X=x_k)$, i.e. the function defining the probabilities of each outcome of the random variable such that:

$$- f(x_k) \geq 0$$

$$- \sum_{k=1}^n f(x_k) = 1$$

PMFs can be visualised graphically, e.g.:

2 D6 PMF



The **Cumulative Distribution Function (CDF)** of a **random variable** X is the function $F(x) = P(X \leq x)$, i.e. the probability that the outcome of X is $\leq$ some value x , defined as:

$$F(x) = \sum_{x_k \leq x} P(X=x_k) = \sum_{x_k \leq x} \underbrace{f(x_k)}_{\text{PMF}}$$

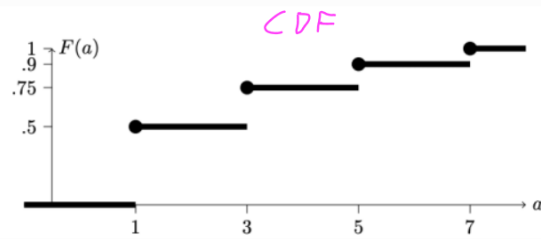
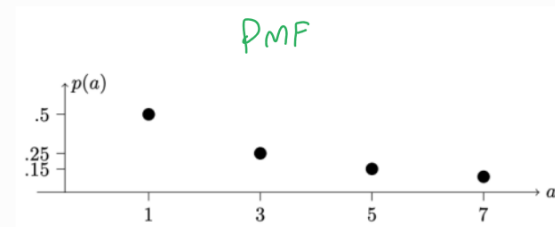
"the sum of the **PMFs** of all possible outcomes of X that are $\leq x$ "

Notes:

- $F(x)$ is always nondecreasing

$$- \lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1$$

All information from the PMF can also be obtained from the CDF.
 → tend to use PMF for discrete random variables as it is not well defined for continuous ones.
 For discrete variables, the CDF graph 'jumps' at each discrete value of the variable:



To summarise a probability distribution of a random variable X , we use the expectation $E(X)$ (mean) and the variance $V(X)$, defined as:

$$E(X) = \sum_x x f(x) = \mu$$

$$V(X) = E((X - E(X))^2) = \sum_x x^2 f(x) - E(X)^2 = \sigma^2$$

Note: Standard deviation $\sigma = \sqrt{V(X)}$

Functions of random variables are also random variables

→ If X is some random variable and h an arbitrary function, then $Y = h(X)$ is also a random variable.

Law of the Unconscious Statistician: If X has PMF f and is discrete, taking values $x_1, \dots, x_n$, then

$$E(Y) = E(h(X)) = \sum_{i=1}^n h(x_i) f(x_i)$$

Properties of transformed random variables, where a and b are constants:

$$\Rightarrow E(aX + b) = aE(X) + b$$

$$\Rightarrow V(aX + b) = a^2 V(X)$$

$$\Rightarrow V(X) = E(X^2) - E(X)^2$$

Two random variables X and Y are independent if, for any sets of outcomes A and B ,

$$P((X \in A) \cap (Y \in B)) = P(X \in A) \times P(Y \in B)$$

For multiple random variables $x_1, \dots, x_n$:

$$E(x_1 + \dots + x_n) = E(x_1) + \dots + E(x_n)$$

If they are all independent:

$$V(x_1 + \dots + x_n) = V(x_1) + \dots + V(x_n)$$

Common Discrete Probability Distributions

Formula sheet:

PMF	Expectation	Variance
Bernoulli: $X \sim \text{Ber}(p)$ $P(X=1) = p$	$E[X] = p$	$V(X) = p(1-p)$
Binomial: $X \sim \text{Bin}(n, p)$ $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	$E[X] = np$	$V(X) = np(1-p)$
Geometric: $X \sim \text{Geom}(p)$ $P(X=k) = p(1-p)^{k-1}$	$E[X] = \frac{1}{p}$	$V(X) = \frac{1-p}{p^2}$
Negative Binomial: $X \sim \text{NegBin}(r, p)$ $P(X=k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$ if $k \geq r$	$E[X] = \frac{r}{p}$	$V(X) = r \frac{1-p}{p^2}$
Poisson: $X \sim \text{Poisson}(\lambda)$ $P(X=k) = e^{-\lambda} \frac{\lambda^k}{k!}$	$E[X] = \lambda$	$V(X) = \lambda$

Discrete Uniform Distribution:

n possible outcomes, each with equal probability $\frac{1}{n}$

For a discrete random variable X with outcomes $a, a+1, \dots, b-1, b$, then:

$$E(X) = \frac{b+a}{2}$$
$$V(X) = \frac{(b-a+1)^2 - 1}{12}$$

Bernoulli Distribution:

If each outcome of a random experiment has only two possibilities (e.g. Heads or Tails), then this is called a **Bernoulli Trial**.

Let these two outcomes A and B be **Success** and **failure** respectively.

From above, $A \cap B = \emptyset$ and $A \cup B = S$

$$\Downarrow$$
$$P(A) + P(B) = 1$$

Let $P(A) = p$, then $P(B) = P(\bar{A}) = 1-p$

A random variable X satisfying the above is said to be a Bernoulli variable with parameter p , denoted $X \sim \text{Ber}(p)$.

$$E(X) = p$$

$$V(X) = p(1-p)$$

Binomial Distribution:

If a random experiment consists of n independent **Bernoulli Trials** such that the probability of Success of each trial, p , is the same, then the random variable X representing the number of Successful trials is a Binomial variable with parameters n, p , denoted $X \sim \text{Bin}(n, p)$.

The **PMF** of X , representing the probability of k **Successful** Bernoulli trials in the experiment, is given by:

$$P(X=k) = f(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

$$E(X) = np$$

$$V(X) = np(1-p)$$

Geometric Distribution:

If a random experiment consists of a series of independent Bernoulli Trials such that the probability of success p is the same across all trials, then the random variable X denoting the number of trials until (and including) the first success is a Geometric variable with parameter p , denoted $X \sim \text{Geom}(p)$.

Note: If we want the probability of success after k failed trials, use $p = k+1$.

The PMF is given by:

$$P(X=k) = f(k) = p(1-p)^{k-1}$$

$$E(X) = 1/p$$

$$V(X) = \frac{1-p}{p^2}$$

The Geometric Distribution is memoryless - past outcomes have no impact on future ones.

$$\Rightarrow P(X=k+m | X > k) = P(X=m)$$

"the probability of no success until the m^{th} next trial is the same regardless of the outcomes of previous trials"

Negative Binomial Distribution:

Generalisation of the Geometric Distribution, this time X denotes the number of trials until r successes; $X \sim \text{Neg Bin}(r, p)$

$$\text{PMF: } P(X=k) = f(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r} \quad \text{if } k > r, \text{ otherwise } 0$$

$$E(X) = r/p$$

$$V(X) = r \frac{1-p}{p^2}$$

Distribution:



This distribution models the number of occurrences of an event in a given time period, assuming all occurrences are independent of one another. Parameter λ = expected number of occurrences in a given time period, $\lambda > 0$. If X follows this distribution, then it is a Poisson variable, denoted $X \sim \text{Poisson}(\lambda)$.

$$\text{PMF: } P(X=k) = f(k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$$E(X) = \lambda$$

$$V(X) = \lambda$$

Continuous Random Variables

A continuous random variable is one that can take a non-countable (infinite) number of possible values.

Since there are infinite possible values, $P(X=x) = 0$ for any single possible value x . We have to consider intervals of values instead.

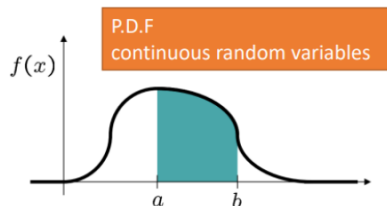
The Probability Density Function (PDF) is used to describe a continuous random variable, denoted f_X or $f(x)$, and is such that:

$$- f(x) > 0 \text{ for all } x$$

$$- \int_{-\infty}^{+\infty} f(x) dx = 1$$

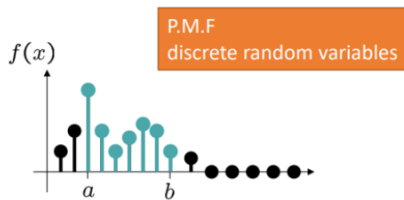
$$- P(a \leq x \leq b) = \int_a^b f(x) dx$$

Note: $P(X < a) = P(X \leq a)$ for continuous variables, since $P(X=a) = 0$.



$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

= area under the curve between a and b



$$P(a \leq X \leq b) = \sum_{x=a}^b f(x)$$

= sum of point masses from a to b

The Cumulative Distribution Function (CDF) of a continuous random variable, denoted $F(x)$, is given by:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

$\Rightarrow t$ is a dummy variable here. We just integrate f from $-\infty$ to x so we cannot use x here.

As with Discrete CDFs:

- $F(x)$ is always nondecreasing
- $\lim_{x \rightarrow -\infty} F(x) = 0$, $\lim_{x \rightarrow +\infty} F(x) = 1$

Also:

$$P(a \leq X \leq b) = F(b) - F(a) \text{ for } a \leq b$$

$$F(x) = F'(x) = \frac{d}{dx} F(x)$$

For a continuous random variable X :

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx$$

$$\begin{aligned} V(X) &= E((X - E(X))^2) = \int_{-\infty}^{+\infty} (x - E(X))^2 f(x) dx \\ &= E(X^2) - E(X)^2 = \int_{-\infty}^{+\infty} x^2 f(x) dx - E(X)^2 \end{aligned}$$

As with discrete variables, any function of a continuous random variable is also a random variable. The Law of the Unconscious Statistician also applies:

$$E(Y) = E(h(X)) = \int_{-\infty}^{+\infty} h(x) f(x) dx$$

As with Discrete:

$$\Rightarrow E(aX + b) = aE(X) + b$$

$$\Rightarrow V(aX + b) = a^2 V(X)$$

$$\Rightarrow V(X) = E(X^2) - E(X)^2$$

Common Continuous Probability Distributions

Formula Sheet:

PDF	Expectation	Variance
Uniform: $X \sim \text{Uniform}([a, b])$ $f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$	$E[X] = \frac{a+b}{2}$	$V(X) = \frac{(b-a)^2}{12}$
Exponential: $X \sim \text{Exp}(\lambda)$ $f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$ The CDF is: $F(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	$E[X] = \frac{1}{\lambda}$	$V(X) = \frac{1}{\lambda^2}$
Normal: $X \sim \mathcal{N}(\mu, \sigma^2)$ $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$E[X] = \mu$	$V(X) = \sigma^2$

Continuous Uniform Distribution:

All possible outcomes of a continuous random variable X within some range $[a, b]$ have the same probability.

$$\text{PDF: } f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$E(X) = \frac{b+a}{2}$$

$$V(X) = \frac{(b-a)^2}{12}$$

Exponential Distribution

Similar to the Geometric Distribution for discrete variables, the Exponential Distribution is used to model the probability of an event occurring after a period of time. The rate parameter λ represents the average number of events occurring in a period of one unit of time. A random variable X following this distribution is called an exponential variable, denoted $X \sim \text{Exp}(\lambda)$.

$$\left. \begin{array}{l} \text{PDF: } f(x) = \lambda e^{-\lambda x} \\ \text{CDF: } F(x) = 1 - e^{-\lambda x} \end{array} \right\} x > 0, \text{ otherwise both} = 0.$$

$$E(X) = \frac{1}{\lambda}$$

$$V(X) = \frac{1}{\lambda^2}$$

The Exponential Distribution is also memoryless, i.e. $P(X > t_1 + t_2 | X > t_1) = P(X > t_2)$

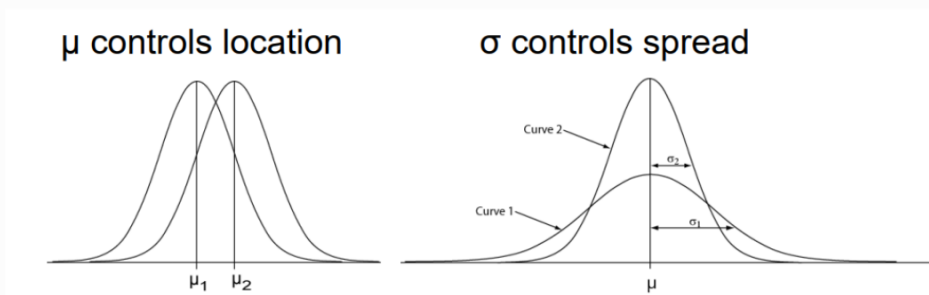
Normal Distribution:

The bell curve distribution. A continuous random variable X with a Normal Distribution with parameters μ and σ^2 , representing the mean and variance respectively, is denoted $X \sim N(\mu, \sigma)$.

$$\text{PDF: } f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$E(X) = \mu$$

$$V(X) = \sigma^2$$



For all values of μ and σ :

$$- P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$$

$$- P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

$$- P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$$

Note: The bell curve of the Normal Distribution is symmetrical about μ

Standard Normal Distribution: a Normal Distribution with $\mu = 0$ and $\sigma = 1$. We use Z to denote random variables with such a distribution (and z for possible outcomes).

$$\text{PMF: } f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

Note: the CDF is denoted $\Phi(z)$ for a Standard Normal variable.

$$P(Z \leq z)$$

To convert a Normally Distributed variable X to a Standard Normally Distributed variable Z :

$$Z = \frac{X - \mu}{\sigma}$$

With the Standard Normal, we can then use the z -table instead of calculating shit manually "
 $\Rightarrow$ all values in the z -table are positive. Use $\Phi(-z) = 1 - \Phi(z)$ to get negative values.

Normal Approximation to Binomial:

If we have a discrete random variable X with a Binomial Distribution, $X \sim \text{Bin}(n, p)$ such that n is large and p is close to 0.5, we can approximate this distribution using a Normal Distribution:

$$X \sim \text{Bin}(n, p) \Rightarrow P(X \leq x) \approx \Phi\left(\frac{x - np}{\sqrt{np(1-p)}}\right)$$

Note: Continuity Correction.

Since Binomial is discrete and Normal is continuous, change:

$$P(X \leq x) \rightarrow P(X \leq x + 0.5)$$

When using this approximation.

This approximation is good for $np > 5$ and $n(1-p) > 5$

Normal Approximation to Poisson

If we have a discrete random variable X with a Fish Distribution, $X \sim \text{Poisson}(\lambda)$, then we can approximate this using the Normal Distribution:

$$X \sim \text{Poisson}(\lambda) \Rightarrow P(X \leq x) \approx \Phi\left(\frac{x - \lambda}{\sqrt{\lambda}}\right)$$

Note: Continuity Correction.

Since Poisson is discrete and Normal is continuous, change:

$$P(X \leq x) \rightarrow P(X \leq x + 0.5)$$

When using this approximation.

This approximation is good for $\lambda > 5$

Multiple Random Variables

We write $P(A \cap B)$ to denote the probability of the intersection of two events A and B . If A and B can be described by random variables X and Y respectively, then we can write:

$$P(A \cap B) = P(X=x \cap Y=y) = P(X=x, Y=y)$$

This is the Joint Probability of $X=x$ and $Y=y$.

Multiple Discrete Random Variables:

The Joint PMF of two discrete random variables X and Y describes how "much" probability each pair of values x, y has, denoted $f(x, y)$ or f_{xy} , equivalent to $P(X=x, Y=y)$

Properties:

$$- f(x, y) \geq 0$$

$$- \sum_x \sum_y f(x, y) = 1$$

The Joint CDF, denoted $F(x, y)$, of two discrete random variables X and Y , is defined as:

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t)$$

Marginal Probability Distributions are used when we only want to look at one variable in a bivariate probability distribution. For a PMF f_{xy} , these are:

$$f_X(x) = P(X=x) = \sum_y P(X=x, Y=y)$$

$$f_Y(y) = P(Y=y) = \sum_x P(X=x, Y=y)$$

$P(X=x, Y=y) = f_{XY}(x, y)$

$y \backslash x$	1	2	3	4	$P(X=x) = f_X(x)$
0	0	0	0	0	0
1	0.10	0.05	0	0	0.15
2	0.05	0.10	0.04	0.01	0.20
3	0	0.05	0.15	0.15	0.35
4	0	0	0.05	0.25	0.30
$P(Y=y) = f_Y(y)$	0.15	0.20	0.24	0.41	

add up to 1

add up to 1

Conditional Distributions are used when we want to calculate conditional probabilities, e.g. $P(Y=y | X=x)$. For a PMF f_{xy} , these are:

$$f_{X|Y=y}(x) = P(X=x | Y=y) = \frac{f_{xy}(x, y)}{f_Y(y)}$$

$$f_{Y|X=x}(y) = P(Y=y | X=x) = \frac{f_{xy}(x, y)}{f_X(x)}$$

For any Set B of outcomes of Y :

$$P(X \in B | Y=y) = \sum_{x \in B} f_{X|Y=y}(x)$$

Joint distributions can be extended to more than 2 variables $x_1, \dots, x_n$, with PMF:

$$f(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n), \quad F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n) = \sum_{a_1 \leq x_1} \dots \sum_{a_n \leq x_n} f(x_1, \dots, x_n)$$

Multiple Continuous Random Variables:

The Joint PDF of two continuous random variables X and Y is denoted $f(x, y)$ or f_{XY} , as the Discrete PMF.

Properties:

$$- f(x, y) \geq 0$$

$$- \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dy dx = 1$$

The Joint CDF of two continuous random variables X and Y , denoted $F(x, y)$, is equivalent to

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy$$

Note: order of integrals does Not matter.

$$\int_{-\infty}^x \int_{-\infty}^y f(x, y) dy dx \text{ is equivalent.}$$

As an extension to this:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

Continuous Marginal Distributions:

$$f_X(x) = P(X=x) = \int_{-\infty}^{+\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = P(Y=y) = \int_{-\infty}^{+\infty} f_{XY}(x, y) dx$$

Continuous Conditional Distributions:

$$f_{X|Y=y}(y) = P(X=x | Y=y) = \frac{f_{XY}(x, y)}{f_Y(y)}$$

- Same as Discrete.

$$f_{Y|X=x}(x) = P(Y=y | X=x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

Similar to Discrete, these can be extended to > 2 variables:

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \dots dx_1$$

Marginal distributions can be obtained by 'integrating out' the other variables, eg-

$$f_{X_7}(x_7) = \int_{a_1}^{b_1} \dots \int_{a_6}^{b_6} \int_{a_8}^{b_8} \dots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \dots dx_8 dx_6 \dots dx_1$$

Independence of Multiple Random Variables:

Two random variables X and Y are independent if one of the following holds:

- $f_X(x) \cdot f_Y(y) = f_{XY}(x,y)$ for all x,y
- $f_{X|Y=y}(x) = f_X(x)$ for all x,y
- $f_{Y|X=x}(y) = f_Y(y)$ for all x,y
- $P(X \in A, Y \in B) = P(X \in A) \cdot P(Y \in B)$ for any sets A and B .

Mean and Variance:

For two random variables X and Y with PMF/PDF $f(x,y)$:

Discrete

$$E(X) = \mu_X = \sum_x x \cdot f_X(x)$$

$$E(Y) = \mu_Y = \sum_y y \cdot f_Y(y)$$

$$V(X) = \sum_x (x^2 \cdot f_X(x)) - E(X)^2$$

$$V(Y) = \sum_y (y^2 \cdot f_Y(y)) - E(Y)^2$$

Continuous - not in slides

$$E(X) = \mu_X = \int_{-\infty}^{+\infty} x \cdot f_X(x) dx$$

$$E(Y) = \mu_Y = \int_{-\infty}^{+\infty} y \cdot f_Y(y) dy$$

$$V(X) = \int_{-\infty}^{+\infty} x^2 \cdot f_X(x) dx - E(X)^2$$

$$V(Y) = \int_{-\infty}^{+\infty} y^2 \cdot f_Y(y) dy - E(Y)^2$$

Properties:

$$\begin{aligned} E(a) &= a \\ E(X+Y) &= \mu_X + \mu_Y \\ E(aX+bY) &= a\mu_X + b\mu_Y \\ E(h(X,Y)) &= \sum_x \sum_y h(x,y) f_{XY}(x,y) \\ V(X) &= E(X^2) - \mu_X^2 \\ V(X+Y) &= \sigma_X^2 + \sigma_Y^2 + 2\sigma_{XY} \\ V(aX+bY+c) &= a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY} \end{aligned}$$

Law of unconscious statistician

Covariance:

The Covariance between two random variables X and Y , denoted $\text{Cov}(X,Y)$ or σ_{XY} , is given by:

$$E(XY) - E(X)E(Y)$$

$$\rightarrow = \sum_x \sum_y xy f(x,y) \text{ (discrete)}$$

$$\text{or } \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x,y) dx dy \text{ (cont.)}$$

Properties:

$$\begin{aligned} \text{Cov}(aX+b, cY+d) &= ac\text{Cov}(X,Y) \\ \text{Cov}(X_1+X_2, Y) &= \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y) \\ \text{Cov}(X, X) &= V(X) \end{aligned}$$

If X and Y independent $\Rightarrow \text{Cov}(X, Y) = 0$

Note: $\text{Cov}(X, Y) = 0 \nRightarrow X$ and Y independent

Correlation Coefficient

The Correlation Coefficient between X and Y , denoted $\text{Corr}(X, Y)$ or ρ_{xy} , is given by:

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{V(X) \cdot V(Y)}}$$

$\text{Corr}(X, Y)$ always lies between -1 and 1 .

Independent $\Rightarrow \text{Corr}(X, Y) = 0$ but not the other way around.

Statistics

Population - the group that includes all 'things' that a researcher is interested in

Sample - a small subset of the population.

Numerical data about the **population** are called **parameters**, eg- μ, σ . The numerical value of these parameters is usually unknown.

Numerical data about a **sample** are called **statistics**. Every statistic is a **random variable**.

For a **population** of size N :

Mean: $\mu = \frac{\sum_{i=1}^N x_i}{N}$

Variance: $\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$

Standard deviation: $\sigma = \sqrt{\sigma^2}$

For a **sample** of size n :

Mean: $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

Variance: $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ Note: -1 for some reason

Standard deviation: $s = \sqrt{s^2}$

The **population distribution** of a variable is the distribution of values of that variable across the population.
The **sampling distribution** of a statistic is the distribution of the values taken by the statistic across all possible samples of a given size of the population.

Markov's Inequality: Given a non-negative random variable X :

$$P(X \geq a) \leq \frac{E(X)}{a} \quad \forall a > 0$$

Chebyshev's Inequality: Given a random variable X , with $E(X) = \mu$, $V(X) = \sigma^2$:

$$P(|X - \mu| \geq K) \leq \frac{\sigma^2}{K^2} \quad \forall K > 0$$

Note: Chebyshev's Inequality can be derived by plugging $X = (X - \mu)^2$, $a = K^2$ into Markov's Inequality.

Distribution of Sample mean:

Let $\bar{X}$ be the mean of a sample of size n taken from a large population. The population has mean μ and variance σ^2 . Then:

$$E(\bar{X}) = \mu$$

$$V(\bar{X}) = \frac{\sigma^2}{n}$$

If the population is normally distributed, then so is $\bar{X}$.

Law of Large Numbers:

Consider n independent, identically distributed random variables $X_1, \dots, X_n$, such that for each X_i , $E(X_i) = \mu$ and $V(X_i) = \sigma^2$. Let $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Then:

Weak Law: for any $\epsilon > 0$, $P(|\bar{X} - \mu| \geq \epsilon) \rightarrow 0$ as $n \rightarrow +\infty$

$n = \text{num. of samples}$

Strong Law: $\lim_{n \rightarrow +\infty} P(\bar{X} = \mu) = 1$

As the sample size, n , increases, the distribution of sample means approaches a Normal Distribution and the mean of this distribution approaches the population mean.

Central Limit Theorem: As $n \rightarrow +\infty$, $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$

$$\Rightarrow Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1) - \text{Standard normal.}$$

$n = \text{Sample size}$

$\rightarrow$ this approximation generally holds for $n > 30$

Confidence Intervals

The Standard Error of a sample statistic is the standard deviation of its sampling distribution.

- For Sample Mean $\bar{X}$, this is $\frac{\sigma}{\sqrt{n}} \rightarrow \text{S.d. of population}$

- For Sample Proportion $\hat{p}$, this is $\sqrt{\frac{p(1-p)}{n}}$ — proportion of population meeting the same condition.

Point Estimation: using a single statistic value to estimate a population parameter.

$\rightarrow$ e.g. $\bar{X}$ estimates μ
 s^2 estimates σ^2
 $\hat{p}$ estimates p

Such statistics are called estimators

The bias $b(\hat{\theta})$ of an estimator $\hat{\theta}$ of a parameter θ is given by:

$$b(\hat{\theta}) = E(\hat{\theta}) - \theta$$

The variance $V(\hat{\theta})$ of an estimator $\hat{\theta}$ of a parameter θ is given by:

$$V(\hat{\theta}) = E[(\hat{\theta} - E(\hat{\theta}))^2]$$

Confidence Interval: using the Standard Error multiplied by a Confidence level with a point estimator to obtain an interval estimate.

$$\text{Confidence Interval} = \text{point estimate} \pm \text{Confidence level} \times \text{Standard error}$$
$$= \hat{\theta} \pm |z_{\alpha}| \sigma$$

z_{α} is the Z-score on a Z-table such that the area under the curve = α

$$\alpha = 1 - \text{Confidence value}$$

$$\Rightarrow 95\% \text{ confidence} \Rightarrow \alpha = 0.05$$

Two-sided Confidence Interval:

$$P(\hat{\theta} - |z_{\alpha/2}| \times \sigma \leq \theta \leq \hat{\theta} + |z_{\alpha/2}| \times \sigma) = 1 - \alpha$$

$\Rightarrow$ we want to find an interval around the point estimate such that the real parameter value has a certain probability $(1 - \alpha)$ of falling within this interval.

One-sided Intervals:

$$P(\hat{\theta} - |z_{\alpha}| \times \sigma \leq \theta) = 1 - \alpha$$

$$P(\theta \leq \hat{\theta} + |z_{\alpha}| \times \sigma) = 1 - \alpha$$

Note: use α instead of $\alpha/2$ here.

Hypothesis Testing

Null Hypothesis H_0 : The known value of a parameter — Conventional wisdom.

Alternative Hypothesis H_1 : the parameter is either $>$, $<$, $\neq$ the value specified in H_0 .

Note: Usually the parameter is either the population mean, μ or the proportion, p , of the population that meet some criteria.

Test Statistics: the value used in making a decision about H_0 .

Mean: $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ $\rightarrow$ value specified in H_0
 $\swarrow$ $\searrow$
Sample mean σ $\rightarrow$ population S.d.
Sample Size

Proportion: $z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$ Note: $\hat{p} = \frac{X}{n}$ so: $z = \frac{X - np}{\sqrt{np(1-p)}}$

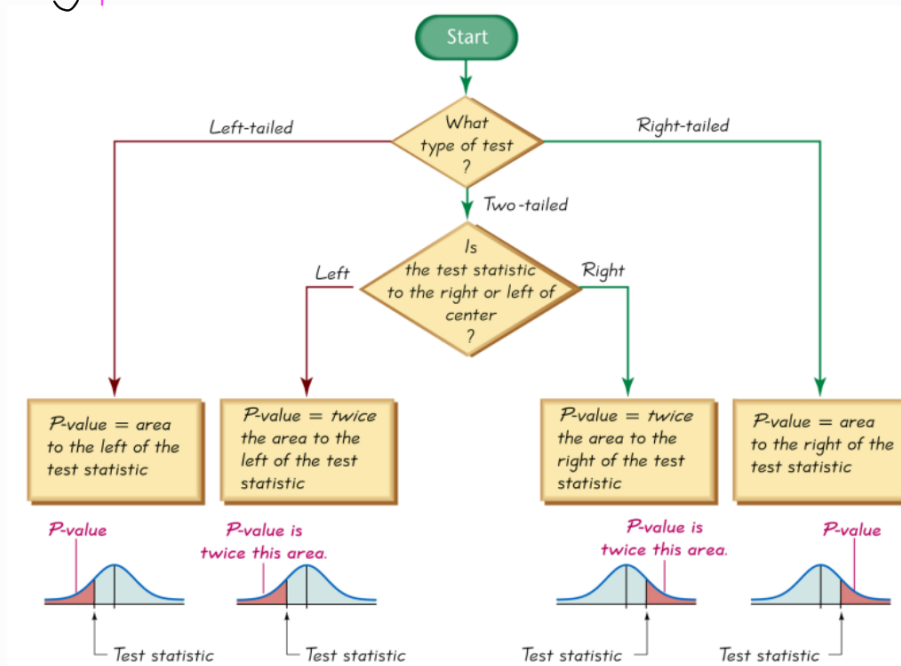
Making a Decision:

P-value method:

P-value: The probability that a sample statistic is $\geq / \leq$ the known value specified in H_0 .

Significance Level: If the p-value is $\leq$ than the Significance Level, reject H_0 in favour of H_1 .
Otherwise, insufficient evidence to reject.

Finding p-values:



Critical Region Method:

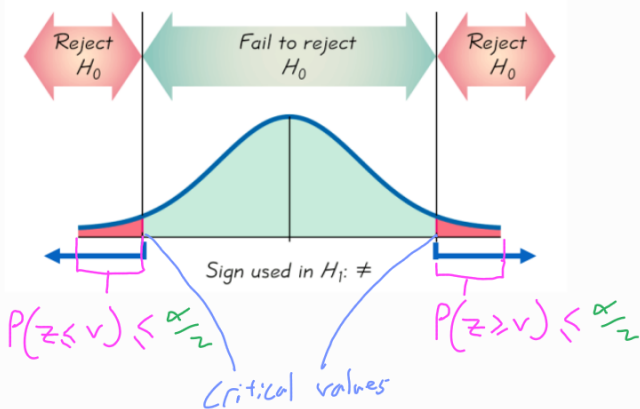
Critical Region: the set of all values of the test statistic that would lead to the rejection of H_0

Critical values: any value inside the critical region that separates the critical region from the values of the test statistic that do not lead to rejection of H_0 .

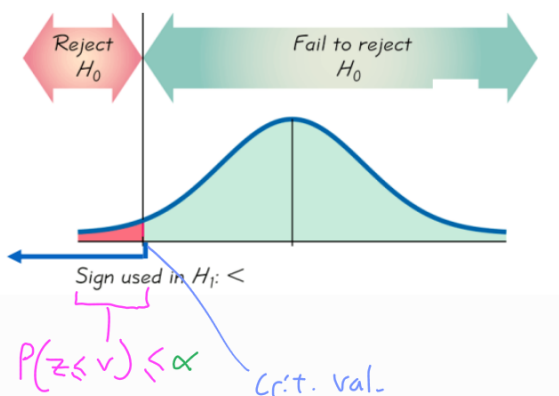
$\Rightarrow$ the first value(s) v such that $P(Z \leq / \geq v) \leq \text{significance level}$

depends on
context of H_1

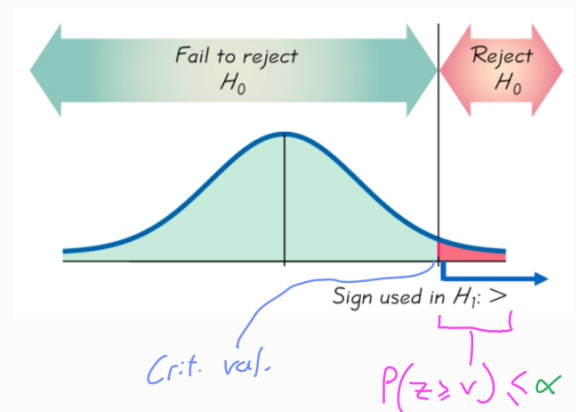
Two-tailed:



Left-tailed:



Right-tailed:



Testing Hypotheses on the Mean, Variance Known (Z-Tests)

Null hypothesis: $H_0: \mu = \mu_0$

Test statistic: $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

Alternative Hypotheses	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: \mu \neq \mu_0$	Probability above $ z_0 $ and probability below $- z_0 $, $P = 2[1 - \Phi(z_0)]$	$z_0 > z_{\alpha/2} $ or $z_0 < - z_{\alpha/2} $
$H_1: \mu > \mu_0$	Probability above z_0 , $P = 1 - \Phi(z_0)$	$z_0 > z_{\alpha} $
$H_1: \mu < \mu_0$	Probability below z_0 , $P = \Phi(z_0)$	$z_0 < - z_{\alpha} $

Testing Hypotheses on a Binomial Proportion

Null hypotheses: $H_0: p = p_0$

Test statistic: $Z_0 = \frac{X - np_0}{\sqrt{np_0(1 - p_0)}}$

Alternative Hypotheses	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: p \neq p_0$	Probability above $ z_0 $ and probability below $- z_0 $, $P = 2[1 - \Phi(z_0)]$	$z_0 > z_{\alpha/2} $ or $z_0 < - z_{\alpha/2} $
$H_1: p > p_0$	Probability above z_0 , $P = 1 - \Phi(z_0)$	$z_0 > z_{\alpha} $
$H_1: p < p_0$	Probability below z_0 , $P = \Phi(z_0)$	$z_0 < - z_{\alpha} $

→ If population variance not known, need to use t-tests. These use the same Test Statistic, only difference is the values come from a different table.

Testing Hypotheses on the Mean of a Normal Distribution, Variance Unknown

Null hypothesis: $H_0: \mu = \mu_0$

Test statistic: $T_0 = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$

Alternative Hypotheses	P-Value	Rejection Criterion for Fixed-Level Tests
$H_1: \mu \neq \mu_0$	Probability above t_0 and probability below $-t_0$	$t_0 > t_{\alpha/2, n-1}$ or $t_0 < -t_{\alpha/2, n-1}$
$H_1: \mu > \mu_0$	Probability above t_0	$t_0 > t_{\alpha, n-1}$
$H_1: \mu < \mu_0$	Probability below t_0	$t_0 < -t_{\alpha, n-1}$

Errors and Power of Hypothesis Tests:

A Type I error is where H_0 is rejected by the hypothesis test, but it is actually True. The probability of this occurring is denoted α .

A Type II error is where H_0 is not rejected by the hypothesis test, but it should have been. The probability of this occurring is denoted β .

The power of a hypothesis test is the probability of rejecting H_0 when H_1 is true, i.e. $1 - \beta$.